

UNIT 7

OBJECTS IN MOTION

Learning objectives:

- Describe position using words like left, right, above, and below.
- Explain motion as a change in position.
- Measure distance using standard and metric units.
- Define speed as how far something moves in a certain time.
- Differentiate between push and pull as forces that move objects.
- Explain how force affects motion (speeding up, slowing down, stopping, or changing direction).
- Describe how friction slows things down and provide examples.
- Explain how different surfaces affect friction and movement.

Position

• How can you describe where something is?

Position is the place where something is. You can describe the position of an object by comparing it to something that does not move. You can use words such as in, on, under, next to, left, right, above, below, near, and far to describe position.



◀ The orange fish is to the left of the chest.

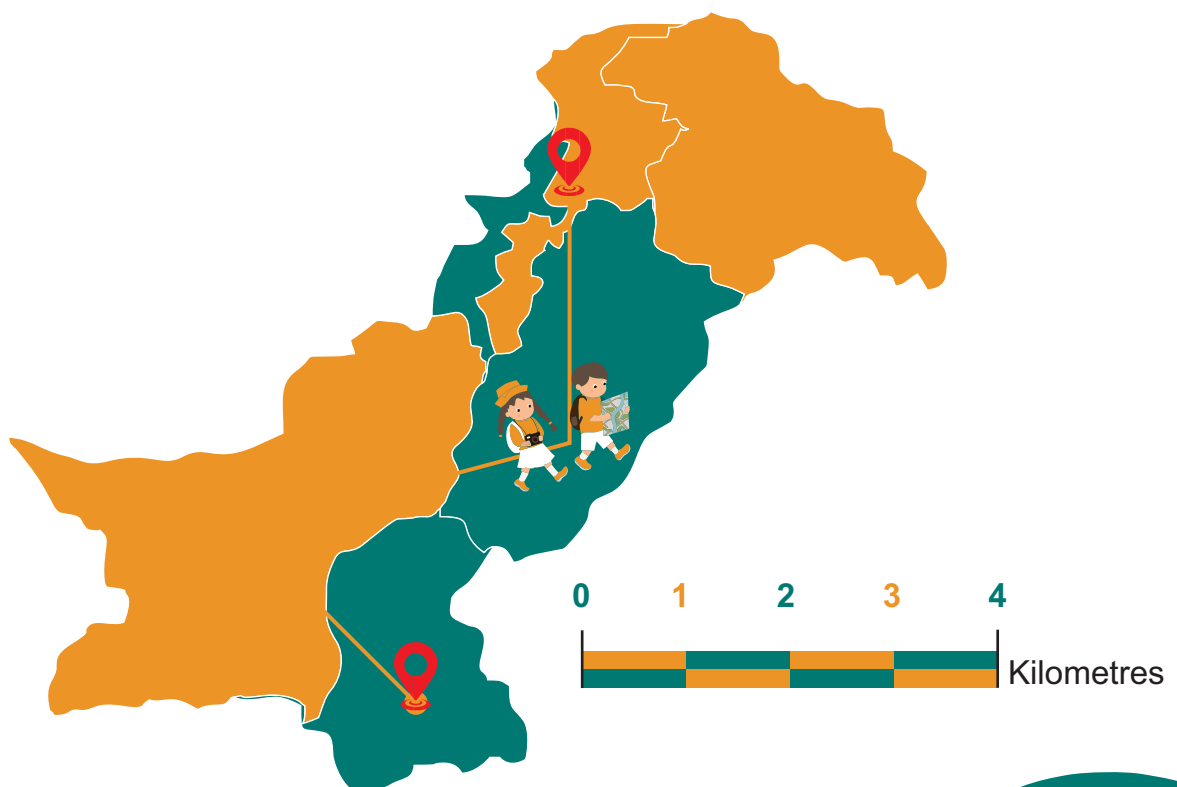
When something moves, it changes its position. You can describe its new position by comparing it to other objects.



▲ **Where is the orange fish now?**
How did it move?

- **How Do You Measure Distance?**

Distance is how far away one thing is from another. There are different ways to measure distance. One way is by using a map.



You can use units, such as inches, feet, and miles to measure distance. You can also use metric units, such as centimetres, metres, and kilometres.



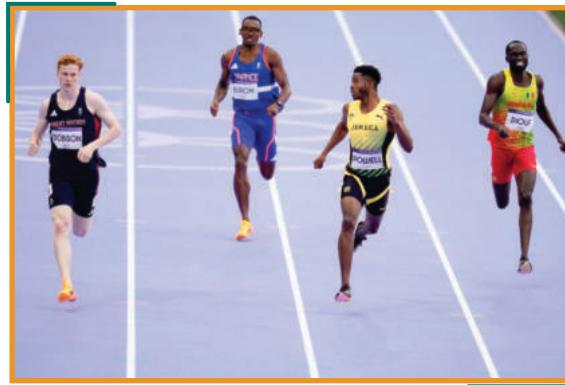
- **How Can You Tell if Something Has Moved?**

All around you, things move. **Motion** is a change in position. When something is moving, we say it is in motion. This diver is in motion. He starts at the top of a diving board. He jumps into the air, and his body changes position.



- **What Is Speed?**

Speed is how far something moves in a certain amount of time. Have you ever watched a race? Most people can run half a mile in 5 minutes. But an Olympic runner can run 1 mile in just 5 minutes or less.



Pushes and Pulls

- **What Are Forces?**

Things cannot move by themselves. You have to use a **force** to make something move. When you play soccer, you kick the ball to move it across the field. Your kick is a force. If you did not kick the ball, it would stay in the same place.



If you **push** something, it moves away from you. A kick is a kind of push. If you **pull** something, it moves closer to you. When you open a drawer, you pull it toward yourself. A **push or a pull** is a force.



▲ The man is pushing the man and boy in the wagon.



▲ In this game, the children on each side of the rope are pulling it towards them.

● What Happens When a Force Changes?

When you use a lot of force to throw a ball, it goes far. When you toss a ball lightly, it will not go as far. If you use more force, things move faster and go further. If you use less force, things move slower and do not go as far.



▲ He will need to use a lot of force to make a basket.



Light things are easier to move than heavy things. You use more force to move something heavy. Some objects are so heavy that people use handcarts, trucks, or cranes to move them.



◀ These basketballs are heavy!

▲ Pulling down on the string makes it easier to move the balls.

Changing Motion

• What Slows Things Down?

Friction is a force that slows down moving things. It happens when two things rub together. There is more friction on a rough surface than on a smooth one. Friction makes it harder to move something on a rough surface than on a smooth one.



▲ To slow down, you drag a rubber stopper on the ground. The dragging causes friction.



Friction can be helpful. Running shoes have treads, or patterns, that add friction. This keeps runners from slipping and falling. Other times, we try to have less friction. The bottoms of ballet slippers are smooth so dancers can slide easily across the floor.



▲ The bottom of a surfboard is smooth so it can glide on the waves.

How a Ball Changes Direction



1. The pitcher uses **force** to throw the ball toward the batter.



2. The batter uses a **push** to hit the ball. It **changes direction** and flies toward the outfield.



- **How Can Forces Change Motion?**

Forces can make things change motion. They can make things speed up, slow down, stop, and start moving. Forces can also make things change direction. In a softball game, the players use force to change the direction of a ball's motion.



3. The player in the field catches the ball. He stops its motion.



Use the clues to find the words. They go across and down.

1. How far away one thing is from another.
2. What is needed to move something.
3. When something changes position.
4. A force that slows down moving things.
5. The place where something is.
6. Move something closer to you.
7. How far something moves in a certain amount of time.
8. Move something away from you.

f	g	x	p	u	l	l	y	n
l	p	e	u	d	a	p	f	r
f	p	o	s	i	t	i	o	n
o	c	u	h	s	f	j	m	p
r	k	q	u	t	z	l	o	g
c	h	n	r	a	v	o	t	s
e	v	x	b	n	d	y	i	z
m	f	r	i	c	t	i	o	n
c	h	s	p	e	e	d	n	a



Forces in Motion

Look at each picture. Circle whether it is an example of push or pull.



PUSH OR PULL



PUSH OR PULL



PUSH OR PULL



PUSH OR PULL



PUSH OR PULL



PUSH OR PULL



Match each action with the type of motion or force it causes. Draw a line to connect them.



Catch a ball

Start moving



Kick a ball

Stop moving



Use your brake

Change direction



Hit a moving ball with bat

Stop moving



Draw arrows to show which direction each object will move when the force is applied.



Test Your Understanding

Q1: What is the difference between motion and position?

Ans: _____

Q2: How do you measure distance between two places?

Ans: _____

Q3: What is speed?

Ans: _____

Q4: Define push and pull.

Ans: _____



Q5: Give an example of a push and a pull in everyday life.

Ans: _____

Q6: What happens when you use more force to push an object?

Ans: _____

Q7: How does friction help runners?

Ans: _____

Q8: Why do dancers wear smooth shoes on stage?

Ans: _____



Fill in the blanks.

1. _____ is a change in position.
2. If you _____ an object, it moves closer to you.
3. Speed is how far something moves in a certain
amount of _____.
4. The more _____ you use, the farther an object
moves.
5. _____ makes it harder to move on a rough
surface.
6. A soccer kick is an example of a _____.



Tick (✓) for True and cross (x) for False statements.

1. Friction helps objects move faster. ()

2. A push moves an object closer to you. ()

3. Speed is measured in units like metres per
second. ()

4. A person pulling a rope is using force. ()

5. A rough surface has less friction than a
smooth one. ()

6. A ball moves farther when hit with greater
force. ()



Choose the correct answer:

1. A car moves faster when you:

- A) Increase friction
- B) Decrease friction
- C) Push it lightly
- D) Pull it back

2. If a runner covers 100 metres in 10 seconds, what is their speed?

- A) 10 metres per second
- B) 100 metres per second
- C) 5 metres per second
- D) 50 metres per second

3. Which of these surfaces will have the most friction?

- A) Ice
- B) Wooden floor
- C) Rough road
- D) Glass

4. A soccer ball stops rolling because of:

- A) Speed
- B) Friction
- C) Position
- D) Distance





5. A person pushing a heavy box applies more force than a person pulling a small chair because:

- A. The chair is heavier
- B) The box needs more force
- C) Pulling is harder than pushing
- D) Distance does not matter





Activity 1



ACTIVITY

Race & Measure: Have students race in different ways (walking, jogging, running) and measure the time taken for a fixed distance. They should compare their speeds and discuss the results.



Activity 2



ACTIVITY

Friction Test: Provide different surfaces (carpet, tile, sandpaper) and let students roll a toy car on each. They record which surface slows the car the most and explain why.

